Moradabad Institute of Technology
Computer Science & Engineering
Sessional Test-1 (Set -1)
Session: 2025-26

Subject Name: Operating System
Subject Code: BCS401
Year: 2nd
Branch: CSE, AIML, DS, IOT

MM: 20
Duration: 1 hr.
Sem: 4th
Section: A, B, C, D, I, J

Q. No	1	2	3	4	5	6
CO	CO1	CO2	CO2	CO1	CO2	CO2
Bloom's Level	K2	K2	K2	K2	K3	K3

CO1	Describe the key components of the operating system, including their management techniques and algorithms
CO2	Explain process and thread management policies, CPU scheduling algorithms, and deadlock handling

Attempt all questions. **Marks**

1) Explain the advantages and limitations of monolithic OS structure. (2)

2) Write and explain the necessary conditions for deadlock to occur. Does cycle in a resource allocation graph is sufficient to confirm deadlock? Explain. (3)

3) Describe different states of a process with the help of a state transition diagram. (3)

4) Explain a) Multiprocessing system (b). Realtime system (4)

5) Consider the following snapshot of a system: (4)

	Allocation			Max			Available		
Process	A	B	C	A	B	C	A	B	C
P0	0	1	0	6	4	2	3	3	2
P1	2	0	0	3	1	1			
P2	3	0	2	7	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	2	3			

(a). What is the content of the matrix need?
(b). Is the system in a safe state?
(c). If a request from process P4 arrives for (4,3,1), can the request be granted immediately?

6) For the processes listed in the following table, which of the following SJF and SRTF scheduling schemes will give the lowest average weighting time? (4)

Process	Arrival Time	Burst Time
P0	0	5
P1	1	3
P2	2	1
P3	3	2

Moradabad Institute of Technology
Computer Science & Engineering
Sessional Test-2 (Set -1)
Session: 2025-26

Subject Name: Operating System
Subject Code: BCS401
Year: 2nd
Branch: CSE. AIML, DS, IOT

MM: 20
Duration: 1 hr.
Sem: 4th
Section: A, B, C, D, I, J

Q. No	1	2	3	4	5	6
CO	CO2	CO4	CO5	CO2	CO4	CO5
Bloom's Level	K1	K2	K2	K3	K3,K4	K3

CO2	Analyze how operating system handles the need for process synchronization and coordination.
CO4	Explain and analyze the memory management and its allocation strategies.
CO5	Determine and assess storage management policies in relation to various storage management technologies.

Note: This paper Contains two sections(A) & (B)
All sections are compulsory.

Section A (08 marks)		
Attempt all questions.		**Marks**
1)	Define semaphores and its types.	(2)
2)	Explain the difference between internal and external fragmentation.	(3)
3)	Describe the various allocation methods of files.	(3)
Section B (12 Marks)		
4)	What is critical section? Write the correct solution to a 2-process critical section problem. Explain how it satisfies mutual exclusion, progress and bounded waiting.	(4)
5)	Explain FIFO and LRU page replacement algorithms with the help of an example. Use the reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 Assume the number of frames = 3. Also, compare the merits and demerits of both algorithms.	(4)
6)	Suppose the head of a moving-head disk with 200 tracks, numbered 0 to 199, is Currently serving a request at track 143 and has just finished a request at track 125. If the queue of requests is kept in FIFO order: 86, 147, 91, 177, 94, 150, 102, 175, 130. What is the total head movement to satisfy these requests for the following Disk scheduling algorithms? (e) SSTF (f) C- SCAN	(4)